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Does gestation vary by ethnic group? A London-based study of over 122 000 pregnancies with spontaneous onset of labour

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Background Evidence exists that normal gestational length varies with ethnicity. This UK-based study compares gestational length amongst a cohort of white European, Black and Asian women.

Methods The cohort comprised 122 415 nulliparous women with singleton live fetuses at the time of spontaneous labour, giving birth in the former North West Thames Health Region, London, UK.

Results The median gestational age at delivery was 39 weeks in Blacks and Asians and 40 weeks in white Europeans. Black women with normal body mass index (BMI) (18.5–24.9 kg/m^2) had increased odds of preterm delivery (odds ratio [OR] = 1.33, 95% CI: 1.15, 1.56, adjusted for deprivation and BMI) compared with white Europeans. The OR of preterm delivery was also increased in Asians compared with white Europeans (OR = 1.45, 95% CI: 1.33, 1.56, adjusted for single unsupported status and smoking). Meconium stained amniotic fluid, which is a sign of fetal maturity, was statistically significantly more frequent in preterm Black and Asian infants and term Black infants compared with white European infants.

Conclusions This research suggests that normal gestational length is shorter in Black and Asian women compared with white European women and that fetal maturation may occur earlier.

Keywords Ethnicity, gestational length, St Mary's Maternity Information system (SMMIS), meconium, neonatal maturity

The estimated date of delivery (EDD) is calculated clinically early in pregnancy, reflecting its social and medical importance. It is calculated using the date of the last menstrual period (LMP) by adding 280 days to the date of the first day of the LMP, giving a point estimate of 40 weeks for gestational length (including the 2 weeks before conception occurs). This method (sometimes known as Naegele's rule, although his method of adding 7 days and subtracting 3 months from the date of the LMP can give a date up to 3 days different to the 280-day method, because of the variation in the length of different months) has to be relied upon in areas without ultrasound access for dating. Even where ultrasound is available, the principle persists of using menstrual dates to determine the EDD, provided the date suggested by ultrasound measurement does not differ by more than 7 days. This is because the use of ultrasound as a dating technique requires the assumption that all fetal measurements are average for the gestation at which they are made, when they may truly be large or small for gestational age at that time. These methods of calculating the EDD are applied regardless of individual medical or demographic characteristics, and do not account for the fact that different babies mature at different rates.

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Few women deliver on their calculated EDD and 5–10% of women deliver preterm.[1,2] Several factors are known to affect the duration of pregnancy, including parity, socio-demographic characteristics, medical complications, previous preterm delivery, cigarette smoking, and maternal age.[2,3]

Obstetric outcomes may differ amongst ethnic groups when managed in the same setting. A British study found significant differences in duration and outcomes of labour when comparing white, Asian, and black women.[4] Shorter gestational length has been observed in certain ethnic groups.[1,5] Two studies have estimated an average gestational length 5 days shorter in black pregnancy.[6,7] One study noted that differences were more strongly associated with the mother's rather than the father's race.[8] Racial differences have also been observed in the rates of preterm (33–37 weeks) and very preterm (<33 weeks) birth in black compared with white women.[9,10] A UK study explored the factors associated with preterm delivery in different ethnic groups and found that gestation was shorter in UK Africans and Afro-Caribbeans even after correction for socioeconomic risk factors.[11]

One hypothesis for shorter average gestational length amongst black infants is that earlier maturation of the feto-placental unit relates to the maternal pelvic size. A smaller pelvis benefits the mother in evolutionary terms in relation to posture and stability when running. However, a smaller pelvis is also associated with a higher incidence of both obstructed labour and maternal mortality. Indeed, Africans have been observed to have amongst the highest emergency caesarean section rates. In fetal terms it is advantageous for the fetus to have a large head because of the improved brain growth. Thus, this creates conflict in the maternal/fetal relationship. It therefore would be in the interest of the fetus to mature faster and deliver earlier to avoid the complications described.

It is well recognized that gross motor skills develop in black infants earlier than in their white counterparts.[12] There is also evidence of earlier fetal maturation. The incidence of the fetal passage of meconium during labour is strongly related to gestational age, increasing from less than 5% at 34 weeks in white European women, to over 25% post EDD.[13] Black infants are significantly more likely to pass meconium *in utero* at all gestational ages, indicating earlier maturation.[14,15]

Perinatal mortality rates also differ amongst ethnic groups. Black infants in the US experience overall higher mortality compared with white infants.[16,17] In the UK, the highest perinatal mortality rates have also been seen in ethnic minorities.[18,19] However, this oversimplifies the relationship between perinatal mortality and ethnicity. Black gestational age specific mortality has been observed to be lower than white infants amongst those born preterm.[17,19] After 37 weeks, this pattern is reversed with higher perinatal mortality amongst black infants compared with white infants.[16] These observations suggest black infants mature earlier compared with white infants hence their survival advantage if born preterm. By contrast, black infants born after 40 weeks gestation may be susceptible to complications of post maturity at earlier gestations than white infants.

The aim of this study was to compare gestational length amongst three ethnic groups in nulliparous women with singleton pregnancies and spontaneous labour.

Materials and Methods

The St Mary's Maternity Information system (SMMIS) database is a maternity database covering 18 of 20 hospitals in the former North West Thames Health Region since 1988. Each participating hospital collects data regarding maternal and neonatal factors on every patient throughout pregnancy. The collected data are sent annually to the Department of Epidemiology and Public Health at St Mary's Hospital. The main data set containing identifiers such as name and postcode is held on a Sun Workstation within the secure environment of the Department also housing the Department of Health Small Area Statistics Unit. To preserve confidentiality, all data used for this analysis were taken from a pseudo-anonymized data set from which all identifiers except the Oracle database number had been removed, thus complying with Section 60 of the Health and Social Care Act 2001. Analysis was performed on the non-attributable data set of 439 425 women who delivered between 1 January 1988 and 31 December 1998. Ethnicity data were self-reported. Women with missing ethnic group data or ethnic groups not in our study were excluded (51 402). Other exclusions were women with multiple pregnancies (4727), ante-partum stillbirths, or stillbirth of indeterminate timing, and induced or spontaneous abortions (964). Only nulliparous women (218 194 multiparous women excluded) and those who laboured spontaneously (excluded 41 723 women with induced or no labour) were included for analysis as previous preterm delivery is a risk factor for subsequent preterm delivery and those induced or who did not labour would not address the study aims.

The white European women were regarded as the reference (control) group. Black African and black Caribbean women were combined into one group, hereafter called Black. The second group comprises women from India, Pakistan, and Bangladesh, hereafter called Asian.

The main outcome measure was gestation at delivery (term birth). This is recorded in whole weeks on the SMMIS database and is derived from the EDD, which is calculated using a combination of LMP (available for 95% women), clinical examination, and ultrasound scan (available for 96% women). Term was defined as ≥37 completed weeks of pregnancy and preterm between 24 and 37 weeks. Length of neonatal stay in intensive care was compared as a binary variable of <1 day or ≥1 day.

Potential confounders were identified before analysis. These were marital status, single unsupported mother status, Carstairs deprivation score, maternal age at delivery, cigarette smoking, maternal height, body mass index (BMI) at booking, gestation at booking, history of diabetes mellitus, history of hypertension, any other ante-natal booking complications, and year of delivery.

Analysis strategy

The two ethnic groups (Black and Asian) under consideration were analysed separately and compared with the reference (white European) group. The distributions of ethnic groups in the cohort were calculated and differences in baseline characteristics were compared with the reference group using χ^2 significance tests.[20] The odds ratio (OR) and 95% CI for the

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association between the ethnic groups and the outcome of interest was calculated. Two-sided likelihood-based significance tests were deemed statistically significant if the associated P-value was $\leq$5%.[21,22] The effect of the identified potential confounders on the unadjusted OR between ethnic group and gestational age at birth was analysed. Information from this analysis was used to develop a log linear logistic regression model. Covariates were added into the model according to their confounding effect and retained if the crude OR changed by over 10% after adjustment. Interaction terms were managed similarly. Statistical significance was assessed using the likelihood ratio test (LRT). Stata 7.0 software (StataCorp, Texas) was used for analysis.

Results

Characteristics of the cohort

There were 122 415 women remaining for analysis after the exclusions above. Of these, 98 370 were white European (80.4%), 7853 (6.4%) were Black, and 16 192 (13.2%) were Asian. Of white European women, 65% were married, compared with 42% of the Black population (χ^2 $P < 0.001$) and 95% of the Asians (χ^2 $P < 0.001$). The Carstairs' deprivation scores indicated greater deprivation amongst Black and Asian groups compared with the white European group (χ^2 $P < 0.001$ for both groups). Smoking was most common amongst white European women (23%) compared with 13% of Black women (χ^2 $P < 0.001$) and only 2% of Asian women (χ^2 $P < 0.001$). The majority of women had a normal BMI (18.5–24.9 kg/m^2). Amongst Black women, 26% were overweight (BMI 25.0–29.9 kg/m^2) and 9% were obese (BMI >30 kg/m^2) (Table 1).

The median gestational age at delivery was 39 completed weeks for the Black and Asian groups but 40 completed weeks for the white European group. Figure 1 shows a left shift in the cumulative frequency curves for gestational age at birth for Black and Asian women compared with white European women, which is apparent from 28 weeks gestation. Of infants born to Black mothers, 0.90% had gestations less that 28 weeks, compared with 0.23% of infants born to white mothers. At 34 weeks 2.69% of infants born to Black mothers had been born compared with 1.26% of infants born to white mothers.

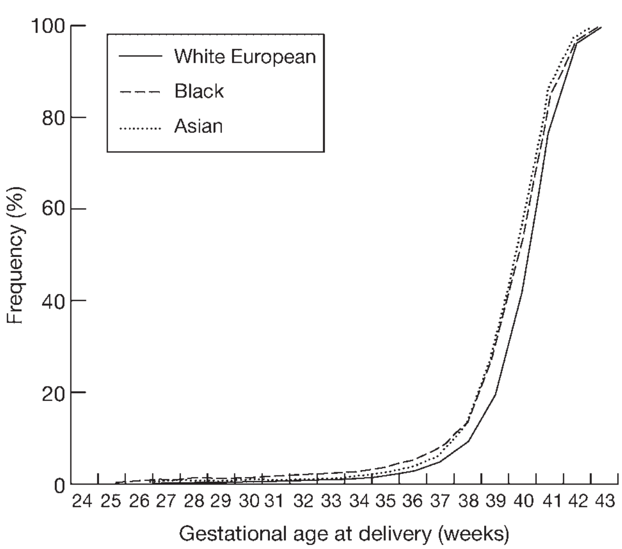

Figure 1 Cumulative frequency chart to show the gestational age at delivery by ethnic group

Risk of preterm delivery in the three ethnic groups

Table 2 presents the risk of preterm delivery overall by maternal age, in the three ethnic groups. Black women overall had the highest proportion of preterm births (7.6%) compared with white Europeans (5.1%) and Asian women (6.5%). In the youngest age group (<20 years) Black women experienced the highest proportion of preterm births.

Risk of term delivery in Black women compared with white European women

Amongst Black women 8% delivered preterm compared with 5% of white European women. The unadjusted odds of preterm delivery in all Black women was significantly higher than in white European women (OR = 1.54, 95% CI: 1.41, 1.69). The unadjusted odds were also calculated for the two sub-groups of the Black group. Black African women had significantly increased odds of preterm delivery (OR = 1.41, 95% CI: 1.23, 1.59) as did black Caribbean women (OR = 1.69, 95% CI: 1.52, 1.92). Further analysis was performed using the combined Black groups. The baseline analysis identified Carstairs' deprivation score and gestation at booking as confounding variables but marital status, cigarette smoking, maternal age, maternal height, booking complications, year of delivery, and single unsupported mother status were not found to be confounders. BMI was an effect modifier (test for homogeneity $P < 0.001$). Carstairs' deprivation score and gestational age at booking were incorporated into a logistic model and the risk of term delivery by ethnic group presented according to different BMI strata. This demonstrated increased odds of Black women with normal BMI of preterm delivery (OR = 1.33, 95% CI: 1.15, 1.56, $P < 0.001$) compared with white European women (Table 3).

Risk of term delivery in Asian women compared with white European women

Amongst Asian women, 7% delivered preterm. The unadjusted odds of preterm delivery in Asian women were significantly higher compared with white European women (OR = 1.30, 95% CI: 1.21, 1.39). Marital status, maternal age, hypertension at booking, other ante-natal booking complications (considered as a binary variable), maternal height, BMI at booking, and year of delivery were found not to confound the relationship between ethnicity and preterm delivery. Cigarette smoking and single unsupported mother status were identified as effect modifiers. The data are thus presented according to smoking and single unsupported mother strata. The association between single unsupported mother status and gestational age at delivery remained significant (LRT $P = 0.04$); in general all results remained statistically significant in the various smoking groups (Table 4).

Neonatal maturity

Length of neonatal stay in a special care unit varied amongst ethnic groups. Overall, all (term and preterm) Black and

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Table 1 The distribution of maternal characteristics by ethnic group

Maternal characteristics		N = 98 370 White European		N = 7853 Black			N = 16 192 Asian		
		N	%	N	%	Chi^2 p[a]	N	%	Chi^2 p[a]
Marital status	**Married**	64 221	65.3	3329	42.4	<0.001	15 363	94.9	<0.001
N = 122 415	**Not married**	34 149	34.71	4524	57.6		829	5.1	
Single unsupported	**No**	76 704	78.0	4763	60.7	<0.001	15 664	96.7	<0.001
mother	**Yes**	21 650	22.0	3083	39.3		528	3.3	
N = 122 392	*Missing (23)*	*(16)*		*(7)*					
Carstairs deprivation	**1**	21 121	23.8	321	4.8	<0.001	1288	8.6	<0.001
score (quintiles)[b]	**2**	21 554	24.3	665	10.0		1771	11.8	
N = 110 333	**3**	22 297	25.2	1228	18.5		3471	23.0	
	4	16 093	18.2	1904	28.7		4114	27.3	
	5	7570	8.5	2511	37.9		4425	29.4	
	Missing (12082)	*(9735)*		*(1224)*			*(1123)*		
Mother's age at	**12–20**	10 014	10.2	1016	12.9	<0.001	1435	8.9	<0.001
delivery (years)	**20–29**	59 221	60.2	5299	67.5		12 093	74.7	
N = 122 405	**30–49**	29 125	29.6	1538	19.6		2664	16.5	
Median 26, IQ range 23–30	*Missing (10)*	*(10)*							
Cigarette smoker	**No**	75 164	76.7	6805	87.0	<0.001	15 809	97.9	<0.001
N = 121 971	**Yes**	22 844	23.3	1015	13.0		334	2.1	
	Missing (444)	*(362)*		*(33)*			*(49)*		
Maternal height (cms)	**≤157.5**	14 843	17.2	1142	17.9	0.131	6893	49.6	<0.001
Median 163,	**>157.5**	71 459	82.8	5224	82.1		6999	50.4	
IQ range158–168	*Missing (15 855)*	*(8123)*		*(870)*			*(1473)*		
N = 106 560									
Body mass index	**Underweight**	2433	2.9	223	3.6	<0.001	1333	10.0	< 0.001
at booking[c]	**Normal**	58 270	69.9	3717	60.7		9381	70.3	
N = 102 825	**Overweight**	17 564	21.1	1620	26.4		2131	16.0	
Median 23, IQ range 21–25	**Obese**	5092	6.1	567	9.3		494	3.7	
	Missing (19 590)	*(15 011)*		*(1726)*			*(2853)*		
Gestation at booking	**≤20 weeks**	80 445	90.5	5360	76.9	<0.001	12 646	83.9	<0.001
N = 110 933	**>20 weeks**	8442	9.5	1611	23.1		2429	16.1	
Median 14, IQ range 12–16	*Missing (11 482)*	*(9483)*		*(882)*			*(1117)*		
History of diabetes	**No**	98 029	99.9	7817	99.9	0.700	16 140	99.9	0.565
mellitus	**Yes**	87	0.1	8	0.1		12	0.1	
N = 122 093	*Missing (322)*	*(254)*		*(28)*			*(40)*		
History of hypertension	**No**	96 695	98.5	7693	98.4	0.18	16 077	99.5	<0.001
N = 122 101	**Yes**	1432	1.5	129	1.7		75	0.5	
	Missing (314)	*(243)*		*(31)*			*(40)*		
Booking complications	**No**	79 665	84.4	5801	82.0	<0.001	12 145	87.1	<0.001
N = 115 384	**Yes**	14 710	15.6	1270	18.0		1793	12.9	
	Missing (7031)	*(3995)*		*(782)*			*(2554)*		
Year of delivery	**1988–89**	19 637	20.0	1393	17.7	<0.001	3212	19.8	<0.001
N = 122 415	**1990–91**	20 467	20.8	1578	20.1		3325	20.5	
	1992–93	19 807	20.1	1786	22.7		3501	21.6	
	1994–95	15 592	15.9	1325	16.9		2546	15.7	
	1996–98	22 867	23.3	1771	22.6		3608	22.3	

[a] Chi^2 test comparing Black and Asian groups (exposures) separately with white Europeans (unexposed).

[b] 1 is least deprived, 5 is most deprived.

[c] Underweight (<18.5 kg/m^2) normal (18.5–24.9 kg/m^2) overweight (25.0–29.9 kg/m^2) obese (>30.0 kg/m^2).

Table 2 Proportion of preterm deliveries by ethnic group and maternal age

	White European		Black		Asian	
	N	%	N	%	N	%
All deliveries	4972	5.1	596	7.6	1049	6.5
Maternal age (years)						
<20	738	14.8	103	17.3	134	12.8
20/29	2789	56.1	365	61.2	722	68.8
≥30	1445	29.1	128	21.5	193	18.4

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Table 3 The estimated odds (OR) ratios for the association between Black ethnicity and preterm delivery

		Total women = 106 223						
		N = 98 370 White European		N = 7853 Black				
		N	%	N	%	OR[a]	95% CI	*P*
Unadjusted:								
Pre-term delivery	**No**	93 398	95.0	7257	92.4			
	Yes	4972	5.1	596	7.6	1.54	1.41–1.69	<0.001
(LRT $P < 0.001$)								
Stratified by BMI and adjusted for Carstairs deprivation score and gestational age at booking:								
BMI [b]	**Underweight**	1950	2.9	179	3.8	0.61	0.28–1.33	0.22
	Normal	47 696	69.7	2816	60.0	1.33	1.15–1.56	<0.001
N = 73 064	**Overweight**	14 517	21.2	1253	26.7	0.87	0.65–1.15	0.32
	Obese	4211	6.2	442	9.4	0.99	0.67–1.49	0.94
(LRT P value ≤0.001)								

[a] Odds ratio of having preterm delivery in Black compared with white European women.
[b] Underweight (<18.5 kg/m^2) normal (18.5–24.9 kg/m^2) overweight (25.0–29.9 kg/m^2) obese (>30.0 kg/m^2).

Table 4 The estimated odds ratios (OR) for the association between Asian ethnic group and preterm delivery

		Total women = 114 562						
		N = 98 370 White European		N = 16 192 Asian				
		N	%	N	%	OR[a]	95% CI	*P*
Unadjusted:								
Preterm delivery	**No**	93 398	95.0	15 143	93.5			
	Yes	4972	5.1	1049	6.5	1.30	1.21–1.39	<0.001
(LRT $P < 0.001$)								
Stratifying by smoking and single unsupported mother status:								
Non-smoker:								
Single unsupported mother	**No**	62 803	83.6	15 370	97.2	1.45	1.33–1.56	<0.001
N = 90 964	**Yes**	12 352	16.4	439	2.8	1.92	1.41–2.63	<0.001
Smoker:								
Single unsupported mother	**No**	13 644	59.7	247	74.0	1.92	1.26–2.86	0.002
N = 23 172	**Yes**	9194	40.3	87	26.1	2.33	1.28–4.17	0.005

[a] Odds ratio of having preterm delivery in Asian compared with white European women.

Asian neonates had a significantly shorter inpatient stay compared with white European infants (Black OR = 1.31, 95% CI: 1.06, 1.62, $P = 0.014$, Asian OR = 1.19, 95% CI: 1.02, 1.38, $P = 0.023$). (Table 5). Black infants were at increased odds of passing meconium both preterm (OR = 1.55, 95% CI: 1.14, 2.17) and at term (OR = 1.52, 95% CI: 1.44, 1.60) compared with white European infants. The Asian preterm infants were also at increased odds of passing meconium compared with white European infants (OR = 1.48, 95% CI: 1.13, 1.93).

Conclusions

For nulliparous women delivering single infants after spontaneous onset of labour the median gestational age at delivery was 39 completed weeks in the Black and Asian groups and 40 completed weeks in the white European group. This divergence in gestational length was apparent from 28 weeks. Black ethnicity was associated with increased odds of preterm delivery. Amongst Black women with a normal BMI, the odds of delivering a preterm baby was 33% higher than that of white European women. Asian ethnicity was also associated with increased odds of preterm delivery. Asian women who were both smokers and single unsupported mothers had the highest odds of preterm delivery compared with white European women.

SMMIS covers 80% of the population within a geographical area as not all hospitals in the region participate in the SMMIS collaboration. The effect of their exclusion on the results can not be quantified, making the introduction of selection bias

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Table 5 The estimated odds ratios of meconium stained amniotic fluid at birth by ethnic group and time of delivery

Preterm infants:

		Total women = 6610									
		N = 4968 White European		N = 594 Black				N = 1048 Asian			
		N	%	N	%	OR[a]	95% CI	N	%	OR[b]	95% CI
Meconium stained	**No**	4725	95.1	550	92.6			974	92.9		
amniotic fluid	**Yes**	243	4.9	44	7.4	1.55	1.14–2.17	74	7.1	1.48	1.13–1.93

Term infants:

		Total women = 115 778									
		N = 93 384 White European		N = 7256 Black				N = 15 138 Asian			
		N	%	N	%	OR[a]	95% CI	N	%	OR[b]	95% CI
Meconium stained	**No**	74 258	79.5	5217	71.9			11 986	79.2		
amniotic fluid	**Yes**	19 126	20.5	2039	28.1	1.52	1.44–1.60	3152	20.8	1.02	0.98–1.07

[a] Odds of having meconium stained amniotic fluid in Black infants compared with odds in white European infants.

[b] Odds of having meconium stained amniotic fluid in Asian infants compared with odds in white European infants.

possible. In the UK, few women have private obstetric care and this enhances the representativeness of this cohort to the general population. SMMIS is useful for studies considering ethnicity as it is based in urban areas, which reflect the ethnic diversity of the UK. By maintaining a uniform system of data collection SMMIS provides high quality data for research, which has been validated by other studies.[23,24]

Classification of racial or ethnic groupings is problematic and reflected by the inconsistent approach of other studies addressing such issues. SMMIS ethnicity classification is based on Department of Health guidelines, which require self-reporting. The 'missing' group formed 3.3% of the total. This may be missing due to non-collection of data or alternatively represent individuals unwilling to assign themselves to an ethnic group. If this is confined to a particular ethnic group this could bias results; however, given the small size of this group, it is unlikely to have a significant influence.

Calculation of gestational age in the hospitals contributing to SMMIS is by a combination of methods as described previously. Inaccuracies are inherently associated with estimation of EDD using LMP alone, but are reduced by using additional ultrasound scan information.[25] EDD is liable to further error the later the first ultrasound scan. Data were unavailable for this study regarding timing of first scan and could be a source of bias in estimating EDD if ethnic groups book for antenatal care at different times. A recent study in England and Wales found that women from ethnic minorities initiated antenatal care later than white British women.[26]

The results of this research are consistent with other studies and indicate that Black ethnicity is associated with decreased gestational length.[1,6,10,11] There is a paucity of data exploring this research question with British Asian women. One study found minimal effect of maternal characteristics on length of gestation but it was carried out in an area with less than the UK average of ethnic minorities and did not define what was meant by the term Asian.[27] Aveyard *et al.*[11] found an increased risk of preterm delivery for Afro-Caribbean women but not for African women. In the former group, they found that half of the excess risk was associated with marital status and deprivation. Henderson & Kay[5] found a decrease in 'Negro' compared with white pregnancy duration but only included women of low socioeconomic class and used LMP for dating.

The finding that Black infants had increased odds of 1.5 of meconium stained amniotic fluid is comparable to the few published studies in this area.[15] It provides epidemiological evidence concerning fetal maturity to support the hypothesis of earlier maturation. Sedaghatian *et al.*[14] found that meconium stained amniotic fluid varied with ethnicity and was highest in 'East African blacks'. However, the ethnic grouping in this study is flawed as it is not based on standard classification, with African and Asian groups subdivided largely on the basis of skin colour.

If there is a difference in gestational length by ethnic group this could potentially influence a few areas of clinical practice, although further data regarding infant morbidity and mortality by ethnicity would be needed first. For example, steroid injections are administered to mothers before 34 weeks to decrease the risk of respiratory distress syndrome, if delivery is needed. There is evidence to support the safety of steroids although repeated doses may increase adverse maternal outcomes.[28] If Black infants mature earlier, it may be possible to halt steroid treatment earlier without compromising survival.

In ethnic groups with shorter gestation it may be appropriate to utilize different definitions of term, for instance reducing cut-off points by one week. For example, women who deliver preterm are regarded as high risk in subsequent pregnancies, which in turn influences clinical management and may limit patient choice. Black women who deliver at 36 weeks may not warrant inclusion in this category. Elective caesarean section is typically performed at 39 weeks gestation. If this standard is applied to all ethnic groups Black and Asian women may be at higher risk of requiring emergency caesarean section with the extra complications this confers. Thus, consideration could be given to slightly earlier delivery in some groups. There are advantageous social implications of increased accuracy of EDD

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prediction such as facilitating better maternity leave planning and child care arrangements.

The presented results suggest that gestational length may vary by ethnicity. However, it is possible that these results have arisen from confounding by other variables, such as environmental factors. The number of available variables in the dataset permits limited further exploration of this point, although there is little evidence that adjustment for socioeconomic deprivation, BMI, maternal height, cigarette smoking, or marital status has any major effect on our results.

Summary

This research provides evidence that the length of human gestation differs amongst ethnic groups in a heterogeneous maternity population. Black and Asian women may have shorter pregnancy duration compared with white European women. The discrepancy in gestational length could be associated with earlier fetal maturation. To predict accurately EDD maternal factors may need consideration. Recognition that gestational length varies by ethnicity could potentially modify principles of obstetric practice.

KEY MESSAGES

- Obstetric outcomes are recognized to vary by ethnicity.
- Gestational length appears to vary by ethnicity.
- Asian and Black women have shorter gestational lengths compared with white European women.
- This may be related to earlier fetal maturation as evidenced by passage of meconium.

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